Sahaj Bhadja

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Summary

Software engineer with 2+ years building systems software, simulation pipelines, and cross-platform services. Strong in C++/Python, Linux-based development, testing, async workflows, and debugging across demanding environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform data-sharing software with a microservice architecture, supporting consistent data flow, distributed validation workflows, and maintainable backend components across Windows, Linux, and macOS production build environments.
- Optimized API and ETL integrations using Python, SQL, AWS Glue, and S3, improving response time by 30% while strengthening pipeline efficiency, systems coordination, and operational behavior across production workloads.
- Maintained 20+ automated tests and CI pipelines with Jenkins, PyTest, and Mockito, increasing release reliability by 68% while improving debugging, deployment consistency, and defect tracking across distributed software builds.
- Executed daily multi-platform build and test automation across Windows, Linux, and macOS, validating installation, integration, and unit-test flows to surface defects early and support repeatable software releases.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built simulation and model-evaluation pipelines for autonomous driving, processing 100K+ trajectory logs and improving long-horizon planning by 18% through structured experimentation, repeatable testing, and quantitative results analysis.
- Architected a spatial-temporal encoder integrating HD-map geometry with agent context, improving closed-loop planning stability by 15% while evaluating performance through controlled simulation rollouts and defect-oriented iterative validation.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built fault-tolerant asynchronous processing workflows with FastAPI background workers, supporting 1K+ AI jobs through explicit job-state transitions, failure isolation, and dependable execution across concurrent production workloads and releases.
- Designed integration pipelines across FastAPI, SQL Server, PostgreSQL, and cloud storage, processing 5K+ monthly jobs with validation checks, reconciliation logic, and CI/CD-backed releases for dependable software operations.
- Translated recurring platform issues into documented fixes and validation paths, helping teams isolate defects faster, verify release behavior consistently, and maintain stable software operations across changing requirements.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Designed concurrent batch-processing workflows and secure file-ingestion APIs for an AI grading platform, handling 10,000+ daily requests while maintaining stable backend behavior, reliable execution, and structured debugging under load.
- Implemented production backend services on AWS with CI/CD-driven releases, reducing feedback turnaround by 60% while supporting multimodal processing, high-throughput workflows, and dependable API performance under peak usage.

Skills

Languages: C++, Python, Java, C, C#, JavaScript, TypeScript, SQL, Bash, HTML, XML
Platforms: Linux, Windows, macOS, Android, Unix fundamentals, cross-platform development, CLI tooling
Systems: OOD, data structures, algorithms, async processing, microservices, real-time workflows, simulation pipelines, operating systems, job scheduling
Testing & Tools: PyTest, Mockito, unit testing, integration testing, defect tracking, peer reviews, validation checks, Jenkins, GitHub Actions, Docker
Infra & Domains: AWS Glue, S3, SQL Server, PostgreSQL, MySQL, Hydra, W&B, embedded systems concepts, robotics, wireless systems interest, software security

Projects

Selected Projects | *C++, React, MERN, TypeScript*
- Implemented FCFS, SJF, SRT, RR, priority scheduling, and MLQ with C++ logic, dynamic metrics, and Gantt visualization across 1K+ simulated process combinations for repeatable operating-systems scheduling experimentation.
- Implemented consent-based patient-history access, Twilio appointment alerts, Sentry monitoring, and scalable TypeScript-backed workflows to reduce administrative workload by 30%, emphasizing reliable state transitions and operational systems visibility.